

1.1 Functions, Domain, and Range - Lesson

MCR3U

Section 1: Relation vs. Function

Definitions

Relation – an identified pattern between two variables that may be represented as a table of values, a graph, or an equation.

Functions – a relation in which each of value of the independent variable (x), corresponds to exactly one value of the dependent variable (y)

Note: All functions are relations but not all relations are functions. For a relation to be a function, there must be only one 'y' value that corresponds to a given 'x' value.

Function or Relation Investigation

1) Complete the following tables of values for each relation:

$$y = x^2$$

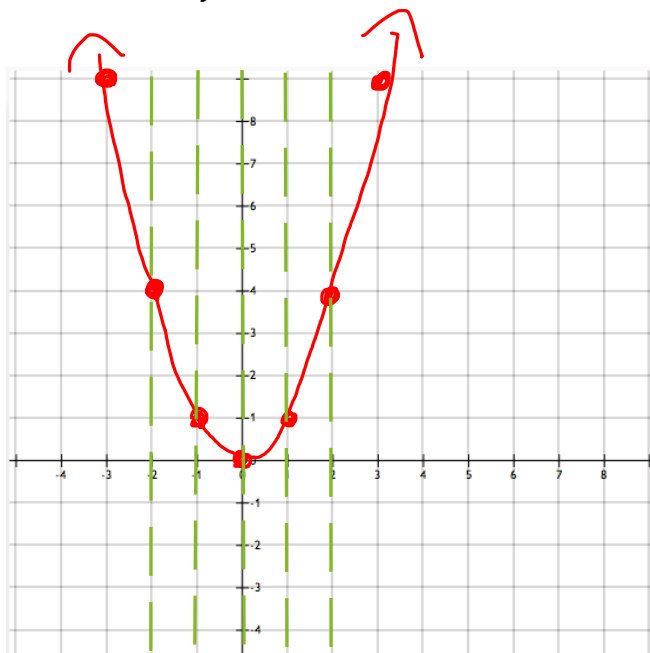
x	y
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9

$$x = y^2$$

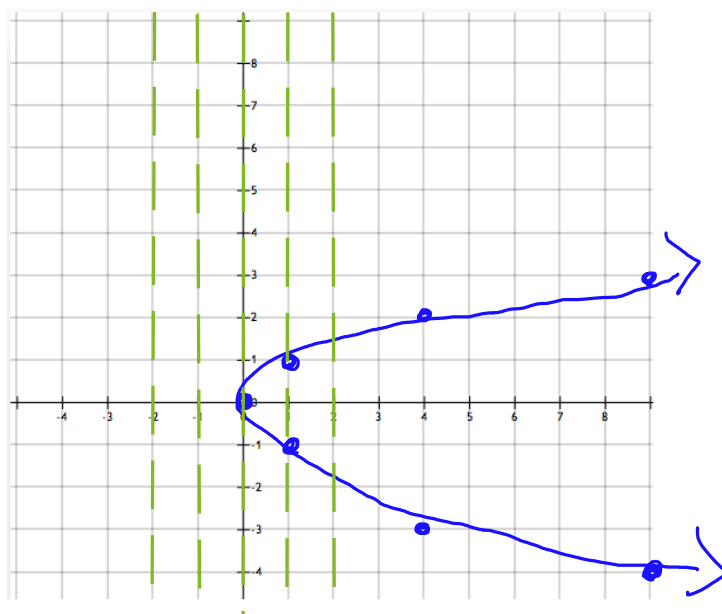
x	y
9	-3
4	-2
1	-1
0	0
1	1
4	2
9	3

2) Graph both relations

$$y = x^2$$



$$x = y^2$$



3) Draw the vertical lines $x = -2$, $x = -1$, $x = 0$, $x = 1$, and $x = 2$ on the graphs above.

4) Compare how the lines drawn in step 3 intersect each of the relations. Which relation is a function? Explain why.

For $y = x^2$, none of the vertical lines drawn intersect the graph at more than one point. That means that for each value of x , there is only 1 corresponding value of y . This means it is a function.

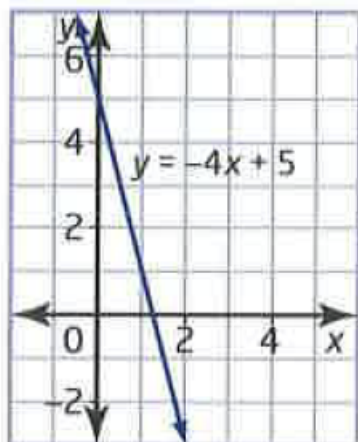
For $x = y^2$, some of the vertical lines drawn intersect the graph at more than one point. That means that some x -values correspond to more than one y -value. This means it is NOT a function.

Section 2: Vertical Line Test

Vertical line test: a method for determining if a relation is a function or not. If every possible vertical line intersects the graph of the relation at only one point, then the relation is a function.

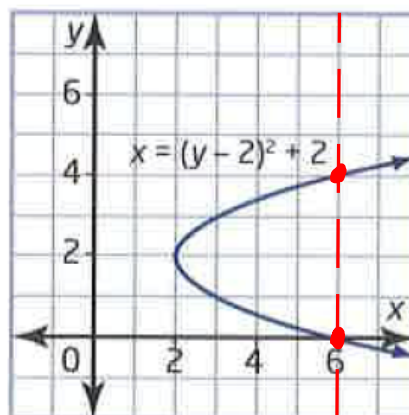
Example 1: Use the vertical line test to determine whether each relation is a function or not.

a)



Function

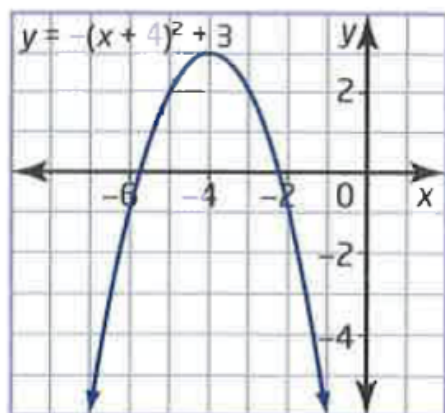
b)



Not a function

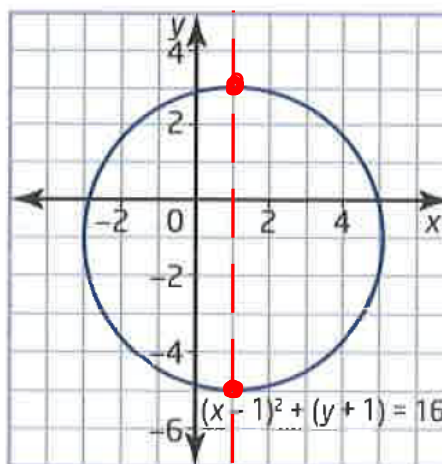
When $x = 6$, $y = 0$ and 4

c)



Function

d)



Not a function

When $x = 1$, $y = -5$ and 3

Section 3: Domain and Range

For any relation, the set of values of the independent variable (often the x-values) is called the domain of the relation. The set of the corresponding values of the dependent variable (often the y-values) is called the range of the relation.

Note: For a function, for each given element of the domain there must be exactly one element in the range.

Domain: values x may take

Range: values y may take

General Notation

$$D: \{x \in \mathbb{R} \mid \text{restrictions}\} \quad \text{or} \quad D: \{x = \#, \#, \dots\}$$

$$R: \{y \in \mathbb{R} \mid \text{restrictions}\} \quad \text{or} \quad R: \{y = \#, \#, \dots\}$$

Real number: a number in the set of all integers, terminating decimals, repeating decimals, non-terminating decimals, and non repeating decimals. Represented by the symbol $\mathbb{R}$

Example 2: Determine the domain and range of each relation from the data given.

a) $\{(-3, 4), (5, -6), (-2, 7), (5, 3), (6, -8)\}$

$$D: \{x = -3, -2, 5, 6\}$$

$$R: \{y = -8, -6, 3, 4, 7\}$$

b)

Age	Number
4	8
5	12
6	5
7	22
8	14
9	9
10	11

$$D: \{x = 4, 5, 6, 7, 8, 9, 10\}$$

$$R: \{y = 5, 8, 9, 11, 12, 14, 22\}$$

Are each of these relations functions?

part a) is NOT a function. There are multiple y-values that correspond to an x-value of 5

part b) is a function. Each value for x has exactly one value for y.

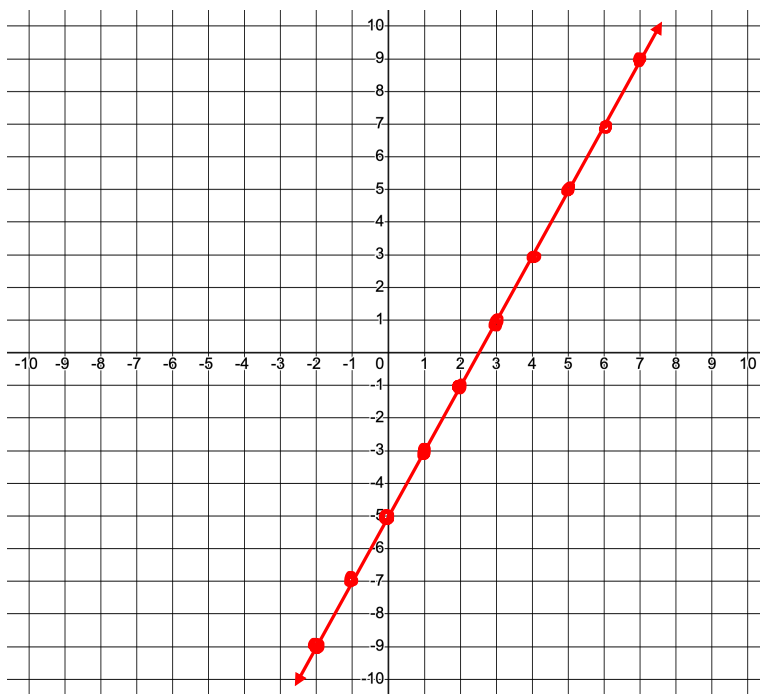
Example 3: Determine the domain and range of each relation. Graph the relation first.

a) $y = 2x - 5$ linear function

↑ slope ↑ y-int

$$D: \{x \in \mathbb{R}\}$$

$$R: \{y \in \mathbb{R}\}$$



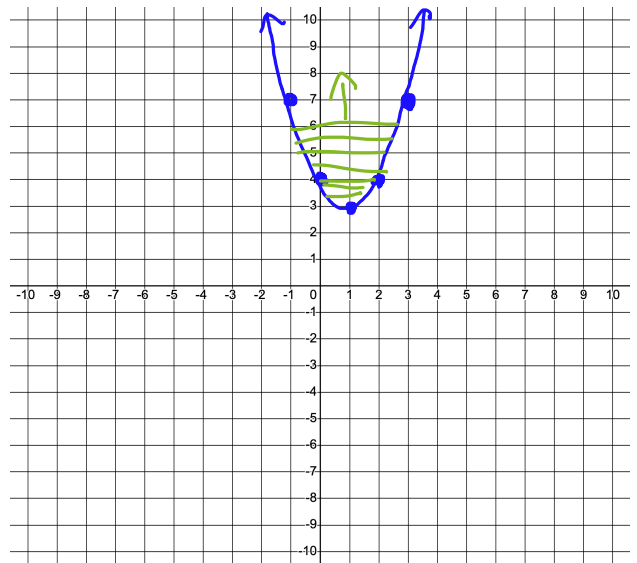
b) $y = (x - 1)^2 + 3$ quadratic function

opens up
vertex at (1,3)

x	y
-1	7
0	4
1	3
2	4
3	7

$$D: \{x \in \mathbb{R}\}$$

$$R: \{y \in \mathbb{R} \mid y \geq 3\}$$

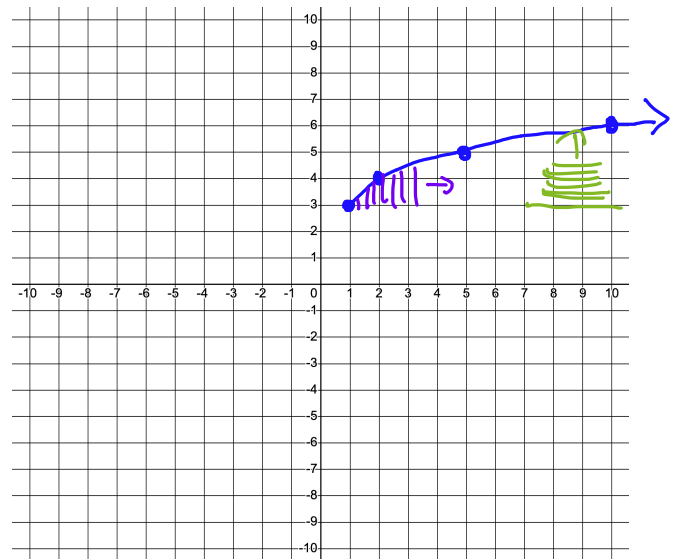


c) $y = \sqrt{x - 1} + 3$ radical function

x	y
1	3
2	4
5	5
10	6

$$D: \{x \in \mathbb{R} \mid x \geq 1\}$$

$$R: \{y \in \mathbb{R} \mid y \geq 3\}$$

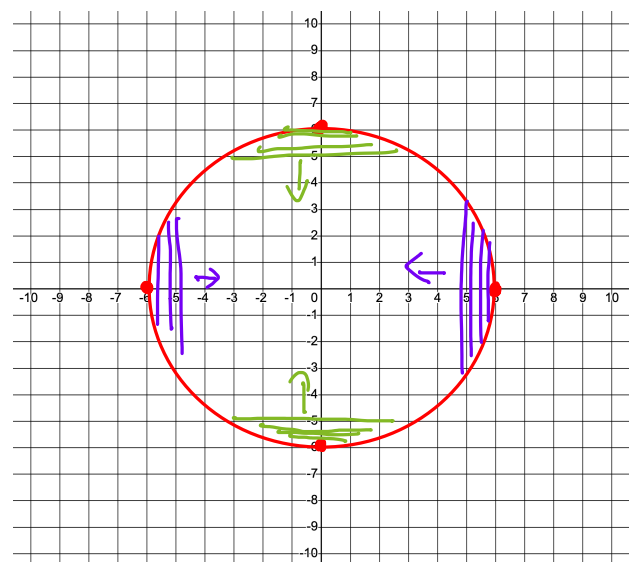


d) $x^2 + y^2 = 36$

circle centered at the origin with
a radius of 6.

$$D: \{x \in \mathbb{R} \mid -6 \leq x \leq 6\}$$

$$R: \{y \in \mathbb{R} \mid -6 \leq y \leq 6\}$$



e) $y = \frac{1}{x+3}$

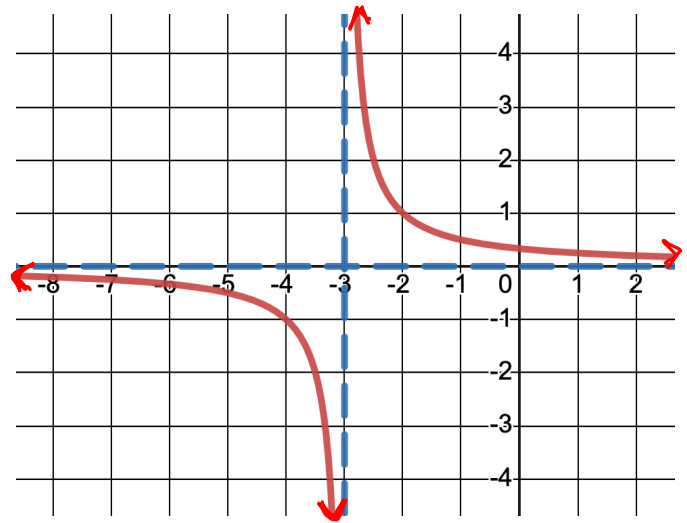
rational function

horizontal asymptote at $y = 0$

vertical asymptote at $x = -3$

$$D: \{x \in \mathbb{R} \mid x \neq -3\}$$

$$R: \{y \in \mathbb{R} \mid y \neq 0\}$$



Asymptotes

Asymptote:

The function $y = \frac{1}{x+3}$ has two asymptotes:

Vertical Asymptote: Division by zero is undefined. Therefore the expression in the denominator of the function can not be zero. Therefore $x \neq -3$. This is why the vertical line $x = -3$ is an asymptote for this function.

Horizontal Asymptote: For the range, there can never be a situation where the result of the division is zero. Therefore the line $y = 0$ is a horizontal asymptote. For all functions where the denominator is a higher degree than the numerator, there will be a horizontal asymptote at $y = 0$.